

MS 6015: Multivariate Statistical Methods for Business

Reference: Johnson and Wichern: Chapter 12

April 24, 2026

Outline

Introduction to Clustering

Similarity Measures

- Qualitative data

- Jaccard Similarity Coefficient

- Other similarity measures

- Conversion of Similarities and Distances

Similarity Coefficient Based Methods-Hierarchical Clustering Methods

- Single Linkage

- Complete Linkage

- Average Linkage

- Ward's Method

Nonhierarchical Clustering

Clustering Based on Statistical Methods

Motivation for Cluster Analysis

Definition

Cluster analysis groups objects so members of the same **cluster** are more **similar** to each other than to objects in other groups- *without* prior labels or class assignments.

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Reveal hidden structure in high-dimensional data where human intuition fails.

Hypothesis Generation

Identify natural groupings suggesting explanatory variables or causal structure.

Data Reduction

Compress datasets into representative group prototypes for downstream modelling.

Applications

Market segmentation · port hinterland classification · genomics · image segmentation.

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Identify natural groupings suggesting explanatory variables or causal structure.

Key insight

Cluster analysis is **unsupervised** - no response variable, no prior labelling. Structure *emerges from the data itself*.

Data Reduction

Compress datasets into representative group prototypes for downstream modelling.

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Introduction

- ▶ Clustering is an important task in analytics in which the data (customers or entities) is grouped into finite subsets such that each subset is a homogeneous group of entities.
- ▶ Clustering assists data scientists to apply appropriate strategies for different clusters identified through cluster characteristics.

Clustering versus Classification

1. Clustering algorithms are unsupervised learning algorithms (classes are not known a priori) whereas logistic regression and classification tree are supervised learning algorithms (where classes are known a priori in the training data).
2. Clustering is descriptive analytics whereas classification is usually a predictive analytics algorithm.

Objective of Clustering

- ▶ The end objective of clustering is to create heterogeneous subsets (clusters) from the original data set such that each subset is homogeneous within the cluster and identify the characteristics that differentiate the subsets.

Classification of Clusters

1. Non-overlapping clusters: Cluster in which each observation belongs to only one cluster. Non-overlapping clusters are more frequently used clustering techniques in practice.
2. Overlapping clusters: An observation may belong to more than one cluster.
3. Probabilistic clusters: An observation may belong to a cluster according to a probability distribution.
4. Hierarchical clustering: Hierarchical clustering creates subsets of data similar to a tree-like structure in which the root node corresponds to the complete set of data. Branches are created from the root node to split the data into heterogeneous subsets (clusters).

Why Similarity / Distance is Central

Core principle

Every clustering algorithm operates exclusively on a **pairwise distance (or similarity) matrix**. The metric choice determines which clusters emerge, their shape, and outlier sensitivity.

Metric Axioms (required for valid d)

- ▶ **Non-negativity:** $d(x, y) \geq 0$; $d(x, y)=0 \Leftrightarrow x=y$
- ▶ **Symmetry:** $d(x, y) = d(y, x)$
- ▶ **Triangle inequality:** $d(x, z) \leq d(x, y) + d(y, z)$

Why it matters

Metric choice affects cluster *shape*, outlier *sensitivity*, and variable-type *suitability*.

Similarity

- ▶ Most efforts to produce a rather simple group structure from a complex data set require a measure of “closeness,” or “similarity.”
- ▶ There is often a great deal of subjectivity involved in the choice of a similarity measure. Important considerations include the nature of the variables (discrete, continuous, binary), scales of measurement (nominal, ordinal, interval, ratio), and subject matter knowledge.
- ▶ When items (units or cases) are clustered, proximity is usually indicated by some sort of distance.
- ▶ By contrast, variables are usually grouped on the basis of correlation coefficients or like measures of association.

Deciding Distance Measures

Choosing the right distance/similarity measure plays an important role in developing clusters.

- ▶ Euclidean distance is valid only for variables under interval and ratio scales
- ▶ Jaccard coefficient should be used for binary variables.
- ▶ For both quantitative and qualitative variables.
 - ▶ Cosine similarity
 - ▶ Gower's distance

Distance and Dissimilarity Measures

1. Euclidean Distance is one of the frequently used distance measures when the variable is either in interval or ratio scale.
2. Standardized Euclidean Distance: One of major issues with Euclidean distance is that the scale of different attributes within the data can be different.
3. Manhattan Distance (City Block Distance): Euclidean distance may not be appropriate while measuring distance between different locations.
4. Minkowski Distance is the generalized distance measure between two cases in the data set.
5. Jaccard Similarity Coefficient (Jaccard Index): is a measure used when the data is qualitative, especially when attributes can be represented in binary form.

Distance and Dissimilarity Measures

6. Cosine Similarity: Consider two cases X_1 and X_2 with n -attributes $X_1 = (x_{11}, x_{12}, \dots, x_{1n})$ and $X_2 = (x_{21}, x_{22}, \dots, x_{2n})$. Then the cosine similarity between X_1 and X_2 is given by

$$\text{Similarity}(X_1, X_2) = \frac{\sum_{t=1}^n x_{1t}x_{2t}}{\sqrt{\sum_{t=1}^n x_{1t}^2} \sqrt{\sum_{t=1}^n x_{2t}^2}}$$

Distance and Dissimilarity Measures

7. Gower's Similarity Coefficient : can be used when the data has both quantitative and qualitative data.

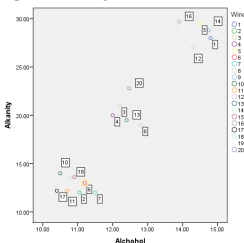
$$D_{ij} = \frac{\sum_{k=1}^n D_{ijk} W_{ijk}}{\sum_{k=1}^n W_{ijk}}$$

Distances and Similarity Coefficients for Pairs of Items

- ▶ Euclidean Measure: $d(x, y) = \sqrt{(x - y)'(x - y)}$
- ▶ Standardized Euclidean Distance: One of major issues with Euclidean distance is that the scale of different attributes within the data can be different.
- ▶ Statistical Distance: $d(x, y) = \sqrt{(x - y)' S^{-1} (x - y)}$
- ▶ Minkowski's Metric: $d(x, y) = \left[\sum_{i=1}^p |x_i - y_i|^m \right]^{1/m}$
- ▶ Canberra Metric: $d(x, y) = \sum_{i=1}^p \frac{|x_i - y_i|}{x_i + y_i}$
- ▶ Czekanowski Coefficient.

Wine Data

- ▶ Data has information about 20 wines sold in the market along with their alcohol and alkalinity of ash content.
- ▶ The plot of 20 wines with respect to alcohol and alkalinity of ash shows that there are three clusters and the wines in these clusters are given by



1. Cluster 1 = {2, 6, 7, 10, 11, 15, 17, 18}
2. Cluster 2 = {3, 4, 8, 13, 20}
3. Cluster 3 = {1, 5, 9, 12, 14, 16, 19}

Standardized Euclidean Example

- ▶ One of major issues with Euclidean distance is that the scale of different attributes within the data can be different.
- ▶ Changing scale of variables can affect the relative sizes of distances.
- ▶ Suppose 3 items have the following bivariate measurements.

X_1	X_2
2	5
4	2
7	9

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X_1	X_2
2	5
4	2
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- ▶ Euclidean Distance Matrices: With original data and X_1 multiplied by 100

$$D = \begin{bmatrix} 0 & 3.6 & 6.4 \\ & 0 & 7.6 \\ & & 0 \end{bmatrix}, D = \begin{bmatrix} 0 & 200 & 500 \\ & 0 & 300 \\ & & 0 \end{bmatrix}$$

- ▶ An easier way of handling the potential bias is to standardize the data using the standardized value.
- ▶ Statistical Distance: $d(x, y) = \sqrt{(x - y)' S^{-1} (x - y)}$

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Presence or Absence of Characteristics

- ▶ When items cannot be represented by meaningful p -dimensional measurements, pairs of items are often compared on the basis of the presence or absence of certain characteristics.
- ▶ Similar items have more characteristics in common than do dissimilar items.
- ▶ The presence or absence of a characteristic can be described mathematically by introducing a binary variable, which assumes the value 1 if the characteristic is present and the value 0 if the characteristic is absent.

Example of Binary Variable

	Var				
	1	2	3	4	5
item i	1	0	0	1	1
item j	1	1	0	1	0

- ▶ In this case, there are two 1-1 matches, one 0-0 match, and two mismatches.
- ▶ The squared Euclidean distance, provides a count of the number of mismatches.
- ▶ Suffers from weighting the 1-1 and 0-0 matches equally e.g., two people both read ancient Greek is stronger evidence of similarity than the absence of this capability
- ▶ A large distance corresponds to many mismatches-that is, dissimilar items.

Similarity Measure for Binary Variables

When the variables are binary, the data can again be arranged in the form of a contingency table. This time, however, the variables, rather than the items, delineate the categories. For each pair of variables, there are n items categorized in the table.

Contingency Table

		Item k		Totals
		1	0	
Item i	1	a	b	$a + b$
	0	c	d	$c + d$
Totals		$a + c$	$b + d$	$p = a + b + c + d$

Some Binary Similarity Coefficients

Coefficient	Rationale
1. $\frac{a + d}{p}$	Equal weights for 1-1 matches and 0-0 matches.
2. $\frac{2(a + d)}{2(a + d) + b + c}$	Double weight for 1-1 matches and 0-0 matches.
3. $\frac{a + d}{a + d + 2(b + c)}$	Double weight for unmatched pairs.
4. $\frac{a}{p}$	No 0-0 matches in numerator.
5. $\frac{a}{a + b + c}$	No 0-0 matches in numerator or denominator. (The 0-0 matches are treated as irrelevant.)
6. $\frac{2a}{2a + b + c}$	No 0-0 matches in numerator or denominator. Double weight for 1-1 matches.
7. $\frac{a}{a + 2(b + c)}$	No 0-0 matches in numerator or denominator. Double weight for unmatched pairs.
8. $\frac{a}{b + c}$	Ratio of matches to mismatches with 0-0 matches excluded.

*[p binary variables; see (12-7).]

Example

Suppose five individuals possess the following characteristics:

	Height	Weight	Eye color	Hair color	Handedness	Gender
Individual 1	68 in	140 lb	green	blond	right	female
Individual 2	73 in	185 lb	brown	brown	right	male
Individual 3	67 in	165 lb	blue	blond	right	male
Individual 4	64 in	120 lb	brown	brown	right	female
Individual 5	76 in	210 lb	brown	brown	left	male

Example: Binary Variables

$$X_1 = \begin{cases} 1 & \text{height} \geq 72 \text{ in.} \\ 0 & \text{height} < 72 \text{ in.} \end{cases}$$

$$X_2 = \begin{cases} 1 & \text{weight} \geq 150 \text{ lb} \\ 0 & \text{weight} < 150 \text{ lb} \end{cases}$$

$$X_3 = \begin{cases} 1 & \text{brown eyes} \\ 0 & \text{otherwise} \end{cases}$$

$$X_4 = \begin{cases} 1 & \text{blond hair} \\ 0 & \text{not blond hair} \end{cases}$$

$$X_5 = \begin{cases} 1 & \text{right handed} \\ 0 & \text{left handed} \end{cases}$$

$$X_6 = \begin{cases} 1 & \text{female} \\ 0 & \text{male} \end{cases}$$

Scores and Mismatches : first 2 individuals

The scores for individuals 1 and 2 on the binary variables are
 $p = 6$

		X_1	X_2	X_3	X_4	X_5	X_6
Individual	1	0	0	0	1	1	1
	2	1	1	1	0	1	0

and the number of matches and mismatches are indicated in the two-way array

		Individual 2		
		1	0	Total
Individual 1	1	1	2	3
	0	3	0	3
Totals		4	2	6

Similarity Matrix with Coefficient 1

$$\begin{array}{c} \begin{array}{ccccc} & 1 & 2 & 3 & 4 & 5 \end{array} \\ \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{array} \left[\begin{array}{ccccc} 1 & & & & \\ 1/6 & 1 & & & \\ 4/6 & 3/6 & 1 & & \\ 4/6 & 3/6 & 2/6 & 1 & \\ 0 & 5/6 & 2/6 & 2/6 & 1 \end{array} \right] \end{array}$$

Similarity Matrix with Coefficient 1

$$\begin{array}{c}
 \begin{array}{ccccc}
 & 1 & 2 & 3 & 4 & 5 \\
 1 & \left[\begin{array}{ccccc}
 1 & & & & \\
 1/6 & 1 & & & \\
 4/6 & 3/6 & 1 & & \\
 4/6 & 3/6 & 2/6 & 1 & \\
 0 & 5/6 & 2/6 & 2/6 & 1
 \end{array} \right]
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Based on the magnitudes of the similarity coefficient, we should conclude that individuals 2 and 5 are most similar and individuals 1 and 5 are least similar

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Jaccard similarity coefficient

- ▶ Jaccard similarity coefficient (JSC) or Jaccard index (Real and Vargas, 1996) is a measure used when the data is qualitative, especially when attributes can be represented in binary form.

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- ▶ Jaccard similarity coefficient (JSC) or Jaccard index (Real and Vargas, 1996) is a measure used when the data is qualitative, especially when attributes can be represented in binary form.
- ▶ JSC for two n -dimensional data , X_1 and X_2 , is given by

$$J(X_1, X_2) = \frac{n(X_1 \cap X_2)}{n(X_1 \cup X_2)}$$

where $n(X_1 \cap X_2)$ is the number of attributes that belong to both X_1 and X_2 , $n(X_1 \cup X_2)$ is the number of attributes that belong to either X_1 or X_2

Example

Consider movie DVD purchases made by two customers as given by the following sets:

- ▶ **Customer 1** = {Jungle Book (JB), Iron Man (IM), Kung Fu Panda (KFP), Before Sunrise (BS), Bridge of spies (BoS), Forrest Gump (FG)}
- ▶ **Customer 2** = {Casablanca (C), Jungle Book (JB), Forrest Gump, Iron Man (IM), Kung Fu Panda (KFP), Schindler's List (SL), The God Father (TGF)}

Binary representation:

Movie Title	BS	BoS	C	FG	IM	JB	KFP	SL	TGF
Customer 1	1	1	0	1	1	1	1	0	0
Customer 2	0	0	1	1	1	1	1	1	1

JSC

JSC

$$JSC = \frac{n(X_1 \cap X_2)}{n(x_1 \cup X_2)} = \frac{4}{9} = 0.44$$

- ▶ Higher the Jaccard coefficient, higher the similarity between two observations being compared.
- ▶ The value of JSC lies between 0 and 1.

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Gower's Similarity Coefficient

- ▶ Gower's Similarity Coefficient : can be used when the data has both quantitative and qualitative data.

$$D_{ij} = \frac{\sum_{k=1}^n D_{ijk} W_{ijk}}{\sum_{k=1}^n W_{ijk}}$$

- ▶ D_{ijk} is the similarity between observations (i and j) for k^{th} variable and W_{ijk} is a binary variable that captures whether the similarity calculation between observations is valid for k^{th} variable. $W_{ijk} = 1$ if the similarity calculation is valid, that is, when both the values are known and 0 otherwise. That is, W_{ijk} takes value 1 only when comparisons are possible, otherwise zero (Gower, 1971).

- ▶ The similarity D_{ijk} for ordinal and continuous variables is given by

$$D_{ijk} = 1 - \frac{|X_{ik} - X_{jk}|}{R_k}$$

Here X_{ik} is the value of observation i for variable k and X_{jk} is the value of observation j for variable k . R_k is the range of variable k in the data set. The value of D_{ijk} lies between 0 and 1.

- ▶ For categorical variables

$$D_{ijk} = \begin{cases} 1 & \text{when } X_{ij} = X_{jk} \\ 0 & \text{otherwise} \end{cases}$$

For binary variables

1. If the variable takes value 1 for both observations then $D_{ijk} = W_{ijk} = 1$.
2. If either i^{th} or j^{th} observation takes value 1, then $D_{ijk} = 0, W_{ijk} = 1$.
3. When both i^{th} and j^{th} take values zero, then $D_{ijk} = W_{ijk} = 0$.

Example: Customer data on movie downloads

Cust	Action	Romance	comedy	Scifi	Rating	Mar. sta
1	23	5	15	0	1	0
2	5	18	16	2	5	1
3	25	0	0	15	5	0
4	2	30	15	0	4	1
5	45	0	0	10	5	0

	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$	$k = 6$	Sum
D_{gk}	0.5814	0.5667	0.9375	0.8667	0.0000	0	2.952
W_{gk}	1	1	1	1	1	1	6

Gower's similarity distance between customer 1 and 2 =
 $\frac{2.952}{6} = 0.492$

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Conversion of Similarities and Distances

- ▶ It is always possible to construct similarities from distances.
For example set

$$\tilde{s}_{ik} = \frac{1}{1 + d_{ik}}$$

- ▶ However, distances that must satisfy properties of distance metric and cannot always be constructed from similarities.
 - ▶ This can be done only if the matrix of similarities is nonnegative definite.

$$d_{ik} = \sqrt{2(1 - \tilde{s}_{ik})}, \tilde{s}_{ii} = 1$$

Similarities and Association Measures for Pairs of Variables

In some applications, it is the variables, rather than the items, that must be grouped. Similarity measures for variables often take the form of sample correlation coefficients. Moreover, in some clustering applications, negative correlations are replaced by their absolute values.

When the variables are binary, the data can again be arranged in the form of a contingency table. This time, however, the variables, rather than the items, delineate the categories. For each pair of variables, there are n items categorized in the table.

Contingency Table

		Variable k		Totals
		1	0	
Variable i	1	a	b	$a + b$
	0	c	d	$c + d$
Totals		$a + c$	$b + d$	$n = a + b + c + d$

Product Moment Correlation as a Measure of Similarity

$$r = \frac{ad - bc}{[(a + b)(c + d)(a + c)(b + d)]^{1/2}}$$

- ▶ Related to the chi-square statistic $r^2 = \chi^2/n$ for testing independence
- ▶ For n fixed, large similarity is consistent with presence of dependence.

Example: Measuring the Similarities of Languages

- ▶ At times, inputs to clustering algorithms may be simple frequencies.
- ▶ A first comparison of languages based on the numerals alone.

Table 12.2 Numerals in 11 Languages

English (E)	Norwegian (N)	Danish (Da)	Dutch (Du)	German (G)	French (Fr)	Spanish (Sp)	Italian (I)	Polish (P)	Hungarian (H)	Finnish (Fi)
one	en	en	een	eins	un	uno	uno	jeden	egy	yksi
two	to	to	twee	zwei	deux	dos	due	dwa	ketto	kaksi
three	tre	tre	drie	drei	trois	tres	tre	trzy	harom	kolme
four	fire	fire	vier	vier	quatre	cuatro	quattro	cztery	negy	neljä
five	fem	fem	vijf	funf	cinq	cinco	cinque	piec	ot	viisi
six	seks	seks	zes	sechs	six	seis	sei	szesc	hat	kuusi
seven	sju	syv	zeven	sieben	sept	siete	sette	siedem	het	seitseman
eight	atte	otte	acht	acht	huit	ocho	otto	osiem	nyolc	kahdeksan
nine	ni	ni	negen	neun	neuf	nueve	nove	dziewiec	kilenc	yhdeksan
ten	ti	ti	tien	zehn	dix	diez	dieci	dziesiec	tiz	kymmenen

Concordant First Letters

We call the words for the same number in two different languages concordant if they have the same first letter and discordant if they do not.

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Table 12.3 Concordant First Letters for Numbers in 11 Languages

	E	N	Da	Du	G	Fr	Sp	I	P	H	Fi
E	10										
N	8	10									
Da	8	9	10								
Du	3	5	4	10							
G	4	6	5	5	10						
Fr	4	4	4	1	3	10					
Sp	4	4	5	1	3	8	10				
I	4	4	5	1	3	9	9	10			
P	3	3	4	0	2	5	7	6	10		
H	1	2	2	2	1	0	0	0	0	10	
Fi	1	1	1	1	1	1	1	1	1	2	10

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Sp	4	4	5	1	3	8	10				
I	4	4	5	1	3	9	9	10			
P	3	3	4	0	2	5	7	6	10		
H	1	2	2	2	1	0	0	0	0	10	
Fi	1	1	1	1	1	1	1	1	1	2	10

English, Norwegian, Danish, Dutch, and German seem to form a group. French, Spanish, Italian, and Polish might be grouped together, whereas Hungarian and Finnish appear to stand alone.

Two Approaches in Creating Clusters

The similarity coefficient (SC)(used for binary data) and hierarchical algorithms (used for continuous data) generally follow one of the two approaches in creating clusters

- ▶ Agglomerative clustering begins with each object as a cluster and merges clusters together until the algorithm converges.
- ▶ Divisive clustering begins with a single cluster containing all objects and splits the cluster successively.

Hierarchical Clustering Methods

- ▶ Agglomerative methods start with each observation as a cluster and with each step combine observations to form clusters until there is only one large cluster.
- ▶ Divisive methods begin with one large cluster and proceed to split into smaller clusters items that are most dissimilar.

Hierarchical Clustering: Agglomerative Methods

- ▶ Initially a many clusters as objects
- ▶ The most similar objects are first grouped
- ▶ Initial groups are merged according to their similarities
- ▶ Eventually, all subgroups are fused into a single cluster

Hierarchical Clustering: Divisive Methods

- ▶ Initial single group is divided into two subgroups such that objects in one subgroup are “far from” objects in the other
- ▶ These subgroups are then further divided into dissimilar subgroups.
- ▶ Continues until there are as many subgroups as objects.

Steps in Hierarchical Cluster Analysis, (Romesburg, 1984)

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- ▶ Using the data matrix, compute the values of a similarity matrix coefficient to measure the similarity (dissimilarity) among all pairs of objects.

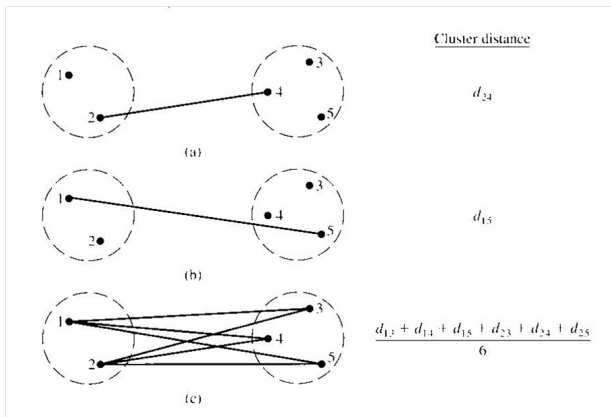
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- ▶ Using the data matrix, compute the values of a similarity matrix coefficient to measure the similarity (dissimilarity) among all pairs of objects.
- ▶ Use a clustering method to process the values of the similarity coefficient, which results in a tree, or dendrogram, that shows the hierarchy of similarities among all pairs of objects. The clusters can be read from the tree.

Different Linkage Methods

- ▶ Single linkage (based on the shortest distance between objects)
- ▶ Complete linkage (based on the largest distance between objects)
- ▶ Average linkage (based on the average distance between objects)
- ▶ Ward's method (based on the sum of squares between the two clusters, summed over all variables), and
- ▶ Centroid method (based on the distance between cluster centroids). Here the distance between clusters is that between their centroids (mean vectors).

Inter-Cluster Distance for Linkage Methods



The Algorithm: Grouping N objects

1. Start with N clusters, each containing a single entity and an symmetric matrix of distances (or similarities) $D = d_k$
2. Search the distance matrix for the nearest (most similar) pair of clusters. Let the distance between “most similar” clusters U and V be d_{UV}
3. Merge clusters U and V . Label the newly formed cluster Update the entries in the distance matrix by (a) deleting the rows and columns corresponding to clusters U and V and (b) adding a row and column giving the distances between cluster (UV) and the remaining clusters.
4. Repeat Steps 2 and 3 a total $N - 1$ of times. (All objects will be in a single cluster after the algorithm terminates.) Record the identity of clusters that are merged and the levels (distances or similarities) at which the mergers take place.

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Nonhierarchical Clustering

Clustering Based on Statistical Methods

Single Linkage or Nearest Neighbour

- ▶ The measure of similarity between two clusters.
- ▶ Distance between “nearest neighbours”
- ▶ At each stage, combine the two clusters that are most similar.
- ▶ $d_{(UV),W} = \min\{d_{UW}, d_{VW}\}$

Example: Clustering using Single Linkage

Consider the hypothetical distances between pairs of five objects as follows:

$$\mathbf{D} = \{d_{ik}\} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 0 & & & & \\ 9 & 0 & & & \\ 3 & 7 & 0 & & \\ 6 & 5 & 9 & 0 & \\ 11 & 10 & 2 & 8 & 0 \end{bmatrix} \end{matrix}$$

$$\begin{array}{c} (35) \quad 1 \quad 2 \quad 4 \\ (35) \left[\begin{array}{cccc} 0 & & & \\ 3 & 0 & & \\ 7 & 9 & 0 & \\ 8 & 6 & 5 & 0 \end{array} \right] \end{array}$$

$$\begin{array}{ccc} (135) & 2 & 4 \\ (135) & \begin{bmatrix} 0 & & \\ 7 & 0 & \\ 6 & 5 & 0 \end{bmatrix} \\ 2 & & \\ 4 & & \end{array}$$

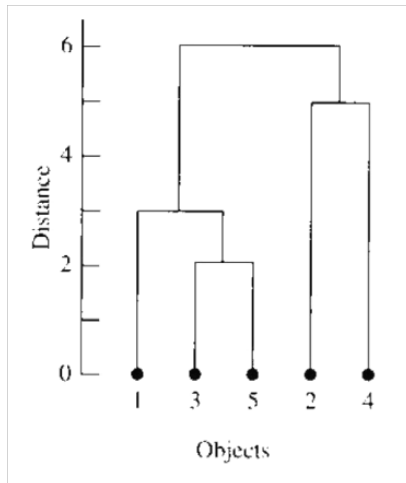
$(135) \quad (24)$

$$\begin{matrix} (135) \\ (24) \end{matrix} \begin{bmatrix} 0 & \\ 6 & 0 \end{bmatrix}$$

Dendrogram

- ▶ Single Linkage Clustering graphically displayed in the form of a dendrogram or tree diagram
- ▶ The branches in the tree represent clusters
- ▶ Branches merge at nodes along a distance axis indicate the level at which fusions occur.

Resultant Dendrogram

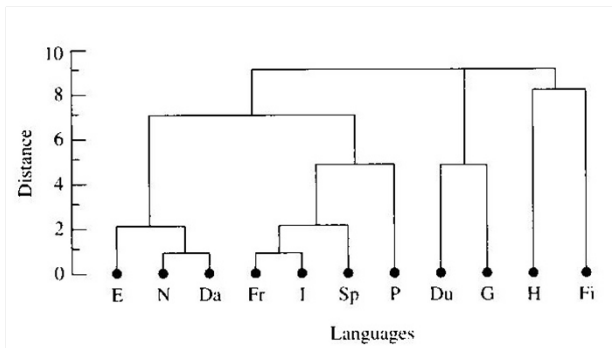


Single Linkage of 11 Languages

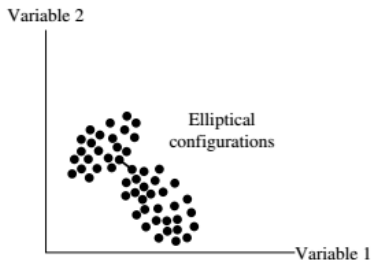
To develop a matrix of distances, we subtract the concordances from the perfect agreement figure of 10 that each language has with itself. The subsequent assignments of distances are

	E	N	Da	Du	G	Fr	Sp	I	P	H	Fi
E	0										
N	2	0									
Da	2	①	0								
Du	7	5	6	0							
G	6	4	5	5	0						
Fr	6	6	6	9	7	0					
Sp	6	6	5	9	7	2	0				
I	6	6	5	9	7	①	①	0			
P	7	7	6	10	8	5	3	4	0		
H	9	8	8	8	9	10	10	10	10	0	
Fi	9	9	9	9	9	9	9	9	9	8	0

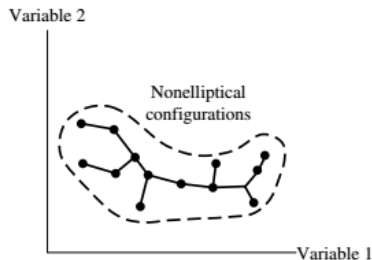
Single Linkage of 11 Languages



Pros and Cons of Single Linkage



(a) Single linkage confused by near overlap



(b) Chaining effect

Outline

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Similarity Measures

Qualitative data

Jaccard Similarity Coefficient

Other similarity measures

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Single Linkage

Complete Linkage

Average Linkage

Ward's Method

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Complete Linkage

$$d_{(UV),W} = \max\{d_{UW}, d_{VW}\}$$

$$\mathbf{D} = \{d_{ik}\} = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 0 & & & & \\ 9 & 0 & & & \\ 3 & 7 & 0 & & \\ 6 & 5 & 9 & 0 & \\ 11 & 10 & 2 & 8 & 0 \end{bmatrix} \end{matrix}$$

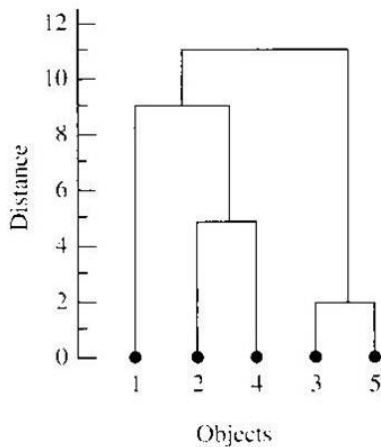
Distance Matrix

$$\begin{array}{c}
 (35) \quad 1 \quad 2 \quad 4 \\
 (35) \begin{bmatrix} 0 & & & \\ 11 & 0 & & \\ 10 & 9 & 0 & \\ 9 & 6 & 5 & 0 \end{bmatrix}
 \end{array}$$

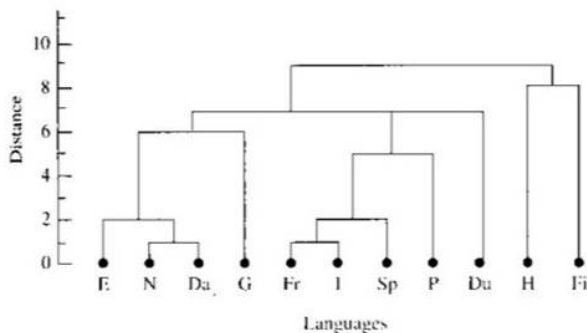
Distance Matrix

$$\begin{array}{ccc} & (35) & (24) & 1 \\ \begin{array}{c} (35) \\ (24) \\ 1 \end{array} & \left[\begin{array}{ccc} 0 & & \\ 10 & 0 & \\ 11 & 9 & 0 \end{array} \right] \end{array}$$

Dendrogram



Complete Linkage of 11 Languages



Clustering variables using complete linkage

- ▶ Data collected on 22 U.S. public utility companies for the year 1975
- ▶ Although it is more interesting to group companies, we shall see here how the complete linkage algorithm can be used to cluster variables.
- ▶ We measure the similarity between pairs of variables by the product-moment correlation coefficient.

Data

Table 12.4 Public Utility Data (1975)

Company	Variables							
	X_1	X_2	X_3	X_4	X_5	X_6	X_7	X_8
1. Arizona Public Service	1.06	9.2	151	54.4	1.6	9077	0.	.628
2. Boston Edison Co.	.89	10.3	202	57.9	2.2	5088	25.3	1.555
3. Central Louisiana Electric Co.	1.43	15.4	113	53.0	3.4	9212	0.	1.058
4. Commonwealth Edison Co.	1.02	11.2	168	56.0	.3	6423	34.3	.700
5. Consolidated Edison Co. (N.Y.)	1.49	8.8	192	51.2	1.0	3300	15.6	2.044
6. Florida Power & Light Co.	1.32	13.5	111	60.0	-2.2	11127	22.5	1.241
7. Hawaiian Electric Co.	1.22	12.2	175	67.6	2.2	7642	0.	1.652
8. Idaho Power Co.	1.10	9.2	245	57.0	3.3	13082	0.	.309
9. Kentucky Utilities Co.	1.34	13.0	168	60.4	7.2	8406	0.	.862
10. Madison Gas & Electric Co.	1.12	12.4	197	53.0	2.7	6455	39.2	.623
11. Nevada Power Co.	.75	7.5	173	51.5	6.5	17441	0.	.768
12. New England Electric Co.	1.13	10.9	178	62.0	3.7	6154	0.	1.897
13. Northern States Power Co.	1.15	12.7	199	53.7	6.4	7179	50.2	.527
14. Oklahoma Gas & Electric Co.	1.09	12.0	96	49.8	1.4	9673	0.	.588
15. Pacific Gas & Electric Co.	.96	7.6	164	62.2	-0.1	6468	.9	1.400
16. Puget Sound Power & Light Co.	1.16	9.9	252	56.0	9.2	15991	0.	.620
17. San Diego Gas & Electric Co.	.76	6.4	136	61.9	9.0	5714	8.3	1.920
18. The Southern Co.	1.05	12.6	150	56.7	2.7	10140	0.	1.108
19. Texas Utilities Co.	1.16	11.7	104	54.0	-2.1	13507	0.	.636
20. Wisconsin Electric Power Co.	1.20	11.8	148	59.9	3.5	7287	41.1	.702
21. United Illuminating Co.	1.04	8.6	204	61.0	3.5	6650	0.	2.116
22. Virginia Electric & Power Co.	1.07	9.3	174	54.3	5.9	10093	26.6	1.306

KEY: X_1 : Fixed-charge coverage ratio (income/debt).
 X_2 : Rate of return on capital.
 X_3 : Cost per KW capacity in place.
 X_4 : Annual load factor.
 X_5 : Peak kWh demand growth from 1974 to 1975.
 X_6 : Sales (kWh use per year).
 X_7 : Percent nuclear.
 X_8 : Total fuel costs (cents per kWh).

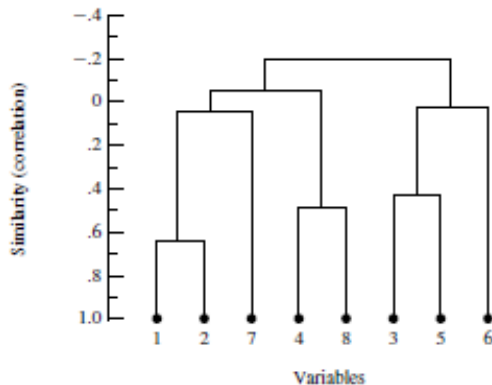
Source: Data courtesy of H. E. Thomson

Correlations Between Pairs of Variables (Public Utility Data)

$$\begin{bmatrix} 1.000 & & & & & & & \\ 0.643 & 1.000 & & & & & & \\ -0.103 & -0.348 & 1.000 & & & & & \\ -0.082 & -0.086 & 0.100 & 1.000 & & & & \\ -0.259 & -0.260 & 0.435 & 0.034 & 1.000 & & & \\ -0.152 & -0.010 & 0.028 & -0.288 & 0.176 & 1.000 & & \\ 0.045 & 0.211 & 0.115 & -0.164 & -0.019 & -0.374 & 1.000 & \\ -0.013 & -0.328 & 0.005 & 0.486 & -0.007 & -0.561 & -0.185 & 1.000 \end{bmatrix}$$

When the sample correlations are used as similarity measures, variables with large negative correlations are regarded as very dissimilar; variables with large positive correlations are regarded as very similar.

Complete Linkage Dendrogram



Interpretation

- ▶ Variables 1 and 2 (fixed-charge coverage ratio and rate of return on capital)
- ▶ Variables 4 and 8 (annual load factor and total fuel costs)
- ▶ Variables 3 and 5 (cost per kilowatt capacity in place and peak kilowatthour demand growth) cluster at intermediate “similarity” levels
- ▶ Variables 7 (percent nuclear) and 6 (sales) remain by themselves until the final stages
- ▶ The final merger brings together the (12478) group and the (356) group.

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Complete Linkage

Average Linkage

Ward's Method

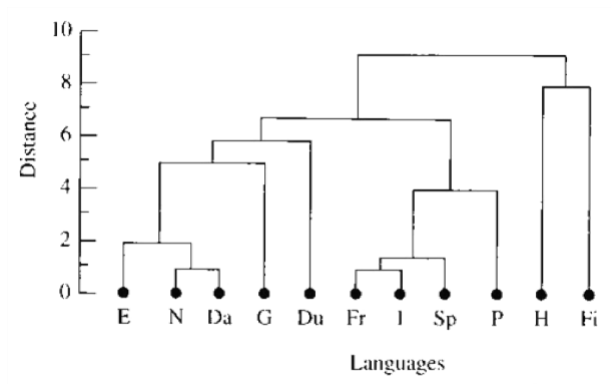
Nonhierarchical Clustering

Clustering Based on Statistical Methods

Average Linkage

Average linkage treats the distance between two clusters as the average distance between all pairs of items where one member of a pair belongs to each cluster.

Average Linkage of 11 Languages



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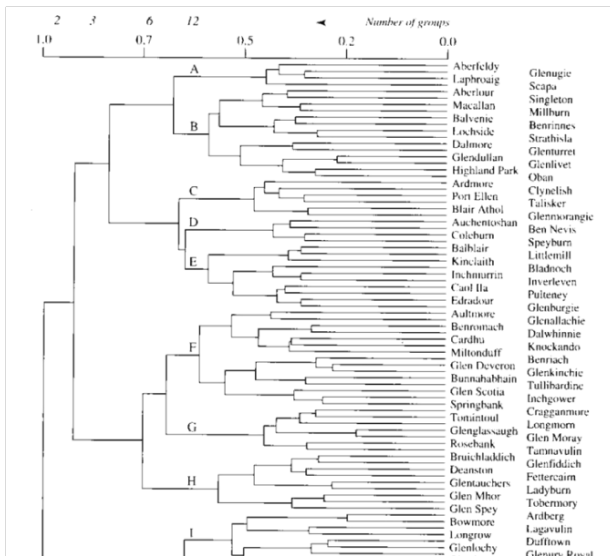
Ward's Hierarchical Clustering Method

- ▶ For a given cluster k , let ESS_k be the sum of the squared deviation of every item in the cluster from the cluster mean

$$ESS = \sum_{j=1}^N (x_j - \bar{x})' (x_j - \bar{x})$$

- ▶ At each step, the union of every possible pair of clusters is considered
- ▶ The two clusters whose combination results in the smallest increase in the sum of ESS_k are joined.

Ward's Clustering Pure Malt Scotch Whiskies

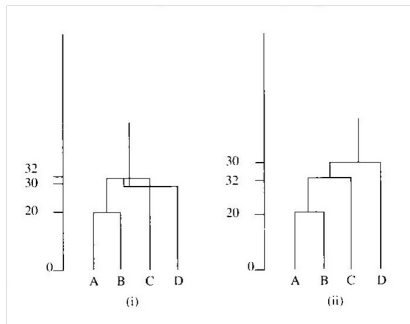


Final Comments : Hierarchical Procedures

- ▶ Sensitive to outliers, or “noise points”
- ▶ No reallocation of objects that may have been “incorrectly” grouped at an early stage.
- ▶ Good idea to try several methods and check if the results are roughly consistent
- ▶ Check stability by perturbation

Inversion

An inversion occurs when an object joins an existing cluster at a smaller distance (greater similarity) than that of a previous consolidation. An inversion is represented two different ways in the following diagram:



Nonhierarchical Clustering K - Means Method

- ▶ Partition the items into K initial clusters
- ▶ Proceed through the list of items, assigning an item to the cluster whose centroid is nearest
- ▶ Recalculate the centroid for the cluster receiving the new item and for the cluster losing the item
- ▶ Repeat until no more reassignment

K-means Objective Function

$$\min_{C_1, \dots, C_K} \sum_{k=1}^K \sum_{x_i \in C_k} \|x_i - \mu_k\|^2$$

Connection to ANOVA

- ▶ ANOVA partitions variance:

$$\text{TSS} = \text{BSS} + \text{WSS}$$

- ▶ K-means minimizes WSS
- ▶ Like ANOVA: more separation between clusters $\Rightarrow$ larger BSS

K-means Variance Decomposition

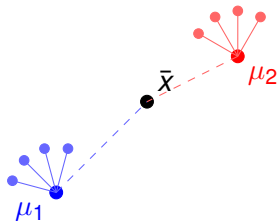
$$\text{TSS} = \sum_i \|x_i - \bar{x}\|^2$$

$$\text{BSS} = \sum_k n_k \|\mu_k - \bar{x}\|^2$$

$$\text{WSS} = \sum_k \sum_{x_i \in C_k} \|x_i - \mu_k\|^2$$

$$\Rightarrow \text{TSS} = \text{BSS} + \text{WSS}$$

Geometric View of K-means



Objective Reformulated via BSS

- ▶ Since TSS is fixed:

$$\min \text{WSS} \iff \max \text{BSS}$$

- ▶ Same principle as ANOVA: maximize explained variance

Illustrative Example: 2D K-means

Dataset:

$(1, 2), (1, 4), (1, 0), (10, 2), (10, 4), (10, 0)$

Initialization: $\mu_1 = (1, 0), \mu_2 = (10, 4)$

Iteration 1: Assignment Step

- ▶ Compute distances to $(1, 0)$ and $(10, 4)$
- ▶ Cluster 1: $(1, 2), (1, 4), (1, 0)$
- ▶ Cluster 2: $(10, 2), (10, 4), (10, 0)$

Iteration 1: Update Step

- Recalculate means:

$$\mu_1 = \left(1, \frac{2 + 4 + 0}{3} \right) = (1, 2)$$

$$\mu_2 = \left(10, \frac{2 + 4 + 0}{3} \right) = (10, 2)$$

Iteration 2: Reassignment Step

- ▶ Assign using new means $(1, 2)$ and $(10, 2)$
- ▶ Same clusters as before - convergence reached

Illustration: Final Clustering

$(1, 4) \bullet$

$(1, 2) \odot$

$(1, 0) \bullet$

$\bullet (10, 4)$

$\odot (10, 2)$

$\bullet (10, 0)$

Illustration: Final Clustering in 2D



Example

Suppose we measure two variables and for each of four items A , B , C , and D . The data are given in the following table

Item	Observations	
	x_1	x_2
A	5	3
B	-1	1
C	1	-2
D	-3	-2

Coordinates of the Centroid

Cluster	Coordinates of centroid	
	$\bar{x}_1$	$\bar{x}_2$
(AB)	$\frac{5 + (-1)}{2} = 2$	$\frac{3 + 1}{2} = 2$
(CD)	$\frac{1 + (-3)}{2} = -1$	$\frac{-2 + (-2)}{2} = -2$

K- Means Method

$$\bar{x}_{i, new} = \frac{n\bar{x}_i + x_{ji}}{n + 1}$$

if the j th item is *added* to a group

$$\bar{x}_{i, new} = \frac{n\bar{x}_i - x_{ji}}{n - 1}$$

if the j th item is *removed* from a group

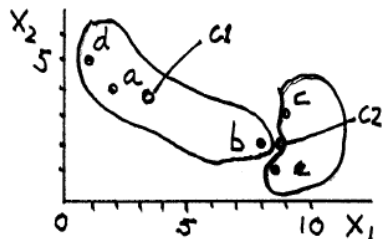
Final Clusters

	Squared distances to group centroids			
	Item			
Cluster	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
<i>A</i>	0	40	41	89
(<i>BCD</i>)	52	4	5	5

Hypothetical Example

Cluster 1			Cluster 2		
Obs.	X_1	X_2	Obs.	X_1	X_2
<i>a</i>	2	4	<i>c</i>	9	3
<i>b</i>	8	2	<i>e</i>	8.5	1
<i>d</i>	1	5			
Ave.	3.67	3.67	Ave.	8.75	2

(a)



(b)

Hypothetical Example

$$D(a, abd) = \sqrt{(2 - 3.67)^2 + (4 - 3.67)^2} = 1.702,$$

$$D(a, ce) = \sqrt{(2 - 8.75)^2 + (4 - 2)^2} = 7.040.$$

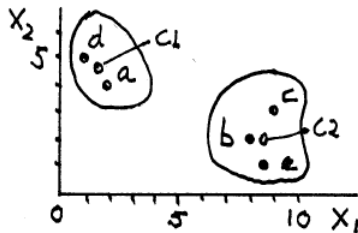
$$D(b, abd) = \sqrt{(8 - 3.67)^2 + (2 - 3.67)^2} = 4.641,$$

$$D(b, ce) = \sqrt{(8 - 8.75)^2 + (2 - 2)^2} = 0.750.$$

Hypothetical Example

Cluster 1			Cluster 2		
Obs.	X_1	X_2	Obs.	X_1	X_2
a	2	4	c	9	3
d	1	5	e	8.5	1
			b	8	2
Ave.	1.5	4.5	Ave.	8.5	2

(a)



(b)

Hypothetical Example

Obs.	Distance from	
	Cluster 1	Cluster 2
<i>a</i>	0.707*	6.801
<i>b</i>	6.964	0.500*
<i>c</i>	7.649	1.118*
<i>d</i>	0.707*	8.078
<i>e</i>	7.826	1.000*

Final Comments-Nonhierarchical Procedures

- ▶ If two or more seed points inadvertently lie within a single cluster, their resulting clusters will be poorly differentiated.
- ▶ The existence of an outlier might produce at least one group with very disperse items.
- ▶ Even if the population is known to consist of K groups, the sampling method may be such that data from the rarest group do not appear in the sample. Forcing the data into K groups would lead to nonsensical clusters.

Summary- Hierarchical versus Nonhierarchical

- ▶ Hierarchical methods
 - ▶ advantage : Do not require a priori knowledge of the number of clusters of the starting partition.
 - ▶ disadvantage : Once an observation is assigned to a cluster it cannot be reassigned to another cluster.
- ▶ Nonhierarchical methods
 - ▶ The cluster centers or the initial partition has to be identified before the technique can proceed to cluster observations-very sensitive to the initial partition.
 - ▶ k-means algorithm and other nonhierarchical clustering algorithms perform poorly when random initial partitions are used. However, their performance is much superior when the results from hierarchical methods are used to form the initial partition.
- ▶ Hierarchical and nonhierarchical techniques should be viewed as complementary clustering techniques rather than as competing techniques.

Clustering Based on Statistical Methods

- ▶ Major advances in clustering methods have been made through the introduction of statistical models that indicate how the collection of $(p \times 1)$ measurements x_j , from the N objects, was generated.
- ▶ The most common model is one where cluster k has expected proportion p_k of the objects and the corresponding measurements are generated by a probability density function $f_k(x)$.

Mixing Distributions

- ▶ If there are K clusters, the observation vector for a single object is modeled as arising from the mixing distribution

$$f_{Mix}(x) = \sum_{i=1}^K p_k f_k(x)$$

where each $p_k \geq 0$ and $\sum_{k=1}^K p_k = 1$.

- ▶ This distribution $f_{Mix}(x)$ is called a mixture of K distributions $f_1(x), f_2(x), \dots, f_K(x)$ because the observations is generated from $f_k(x)$ with probability p_k .

Normal Mixture Model

- ▶ The most common mixture model is a mixture of multivariate normal distributions where the k -th component $f_k(x)$ is the $N_p(\mu_k, \Sigma_k)$ density function.
- ▶ The normal mixture model for one observation x is

$$f_{Mix}(x|\mu_1, \Sigma_1, \dots, \mu_K, \Sigma_K) = \sum_{k=1}^K p_k \frac{1}{(2\pi)^{p/2} |\Sigma_k|^{1/2}} \exp\left(-\frac{1}{2}(x - \mu_k)' \Sigma_k^{-1} (x - \mu_k)\right)$$

Clusters generated by this model are ellipsoidal in shape with the heaviest concentration of observations near the center.

Likelihood Method

Inferences are based on the likelihood, which for N objects and a fixed number of clusters K , is

$$\begin{aligned} L(p_1, \dots, p_K, \mu_1, \Sigma_1, \dots, \mu_K, \Sigma_K) &= \prod_{j=1}^N f_{\text{Mix}}(x | \mu_1, \Sigma_1, \dots, \mu_K, \Sigma_K) \\ &= \prod_{j=1}^N \left(\sum_{k=1}^K p_k \frac{1}{(2\pi)^{p/2} |\Sigma_k|^{1/2}} \exp \left(-\frac{1}{2} (x - \mu_k)' \Sigma_k^{-1} (x - \mu_k) \right) \right) \end{aligned}$$

where the proportions $p_1, \dots, p_K$, the mean vectors $\mu_1, \dots, \mu_K$, and the covariance matrices $\Sigma_1, \dots, \Sigma_K$ are unknown.

Model Selection

- ▶ A good approach to selecting a model is to first obtain the maximum likelihood estimates $\hat{p}_1, \dots, \hat{p}_K, \hat{\mu}_1, \hat{\Sigma}_1, \dots, \hat{\mu}_K, \hat{\Sigma}_K$ for a fixed number of clusters K .
- ▶ The resulting value of the maximum likelihood

$$L_{max} = L(\hat{p}_1, \dots, \hat{p}_K, \hat{\mu}_1, \hat{\Sigma}_1, \dots, \hat{\mu}_K, \hat{\Sigma}_K)$$

provides the basis for model selection.

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provides the basis for model selection.

- ▶ Question: How do we decide on a reasonable value for the number of clusters K ?

Number of Clusters

https://doi.org/10.1142/9789812832153_0010

One of the challenges in clustering is identification of the ideal number of clusters and the quality of clusters.

1. CH Index: $CH(k) = \frac{B(k)/k-1}{W(k)/(n-k)}$
2. Hartigan: $H(k) = \left\{ \frac{W(k)}{W(k+1)} - 1 \right\} / (n - k - 1)$
3. Silhouette statistic: $S(i) = \left(\frac{b(i)-a(i)}{\text{Max}\{a(i), b(i)\}} \right)$
4. Elbow method in which the ideal number of clusters is given by the position of elbow in an L-shaped curve.

Comparison of Models

- ▶ In order to compare models with different numbers of parameters, a penalty is subtracted from twice the maximized value of the log-likelihood to give

$$-2\text{Ln}L_{\max} - \text{Penalty}$$

where the penalty depends on the number of parameters estimated and the number of observations N .

- ▶ Since probabilities p_k sum to 1, there are $K - 1$ probabilities that must be estimated, $K \times p$ means and $K \times p(p + 1)/2$ variances and covariances.

AIC and BIC

- ▶ For the Akaike information criterion (*AIC*), the penalty is $2N \times$ (number of parameters) so

$$AIC = 2\ln L_{max} - 2N \left(K \frac{1}{2} (p+1)(p+2) - 1 \right)$$

- ▶ The Bayesian information criterion (*BIC*) is similar but uses the logarithm of the number of parameters in the penalty function

$$BIC = 2\ln L_{max} - 2\ln(N) \left(K \frac{1}{2} (p+1)(p+2) - 1 \right)$$